Software Requirements Specification

for

AI Powered Pathological Myopia Diagnosis System (MyopiaDx)

Version 1.0 approved

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Table of Contents

1. Introduction

1.1 Purpose

1.1.1 Product Name: AI-Powered Pathological Myopia Diagnosis System (MyopiaDx)

1.1.2 Version: 1.0

1.1.3 Scope: This SRS covers an AI-Powered Pathological Myopia Diagnosis System, focusing on the software component that processes ophthalmic images uploaded by eye specialists from imaging devices, analyzes them, summarizes observations and recommends treatment. The document includes the requirements for the image processing subsystem, analysis algorithms, user interface for specialists, and the integration with imaging devices. It does not cover the imaging hardware or the storage infrastructure.

1.2 Document Conventions

Below are the conventions followed in this document:

- Boldness of a word or sentence shows high emphasis
- Font 12, Times New Roman
- Line spacing 1.5
- Alignment Justified
- The cascaded numbering of sections (e.g 1 => 1.1, 1.2 => 1.1.1, 1.1.2) groups subtopics in their topics for example 1.1 and 1.2 subtopics fall under topic 1.
- Page numbers are placed at the bottom right corner of the pages

1.3 Intended Audience and Reading Suggestions

1.3.1 1.3.1 Intended Audience

This document is intended for the following audiences:

- **Developers**: To understand the technical requirements and specifications for implementation.
- **Project Managers**: To oversee the project's progress and ensure alignment with requirements.
- **Users**: To comprehend the system's functionalities and limitations.
- **Testers**: To design and execute test cases based on the specified requirements.
- **Documentation Writers**: To produce user manuals and other relevant documentation.

1.3.2 Reading suggestions

Readers should start with the "Introduction" for a high-level understanding, followed by the "Overall Description" to get a sense of the product's place and function within the larger ecosystem. Developers and Project Managers will benefit from a detailed dive into the "System Features" and "External Interface Requirements," while Testers should focus on the "Performance Requirements" and "Security Requirements" for creating their test plans.

1.4 Product Scope

1.4.1 Purpose

The purpose of this software is to assist eye specialists by analyzing uploaded ophthalmic images and providing detailed diagnostic observations and thereby providing a second opinion.

1.4.2 1.4.2 Benefits

- Enhanced diagnostic accuracy through AI analysis.
- Reduced time for specialists in diagnosing eye conditions.
- Improved patient outcomes with timely and precise diagnostics.

1.4.3 Objectives

- To provide a reliable and efficient diagnostic tool for eye specialists.
- To integrate seamlessly with existing ophthalmic imaging devices.
- To offer additional functionalities such as predictive analytics, treatment and recommendations.

1.4.4 Goals

- To support eye specialists in delivering high-quality patient care.
- To enhance the efficiency of the diagnostic workflow.
- To ensure the software aligns with corporate goals of innovation in healthcare technology.

1.4.5 References

Kalyanasundaram, A., Prabhakaran, S., Briskilal, J., & Kumar, D. S. (2020). Detection of pathological myopia using convolutional neural networks. International Journal of Psychosocial Rehabilitation, 24(5).

2. Overall Description

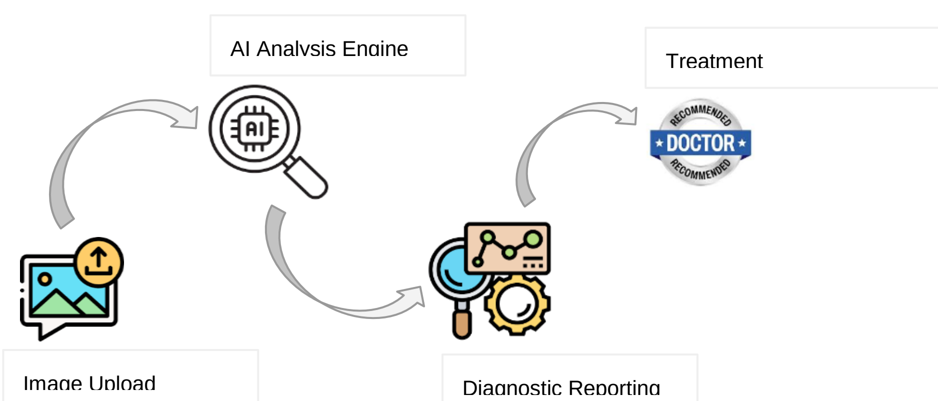
2.1 Product Perspective

2.1.1 Context and Origin

The AI-Powered Diagnosis System for Ophthalmic Imaging will be a new, self-contained product aimed at improving the diagnostic accuracy and efficiency for eye specialists. This product will not be part of an existing product family but rather a novel innovation in medical diagnostics.

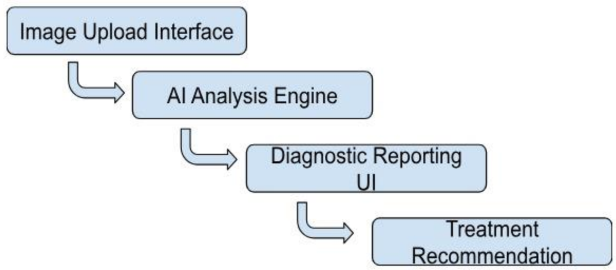
2.1.2 Relation to Larger System

The system will function independently but can be integrated with existing electronic health record (EHR) systems to provide seamless access to patient records. Additionally, it will be able to connect with various ophthalmic imaging devices to streamline the workflow from image capture to diagnosis.



2.2 Product Functions

- Image Upload from imaging devices like fundus cameras.
- Analysis of Ophthalmic Images
- Diagnostic Reporting
- Treatment Recommendation
- Integration with EHR Systems (Patient records systems)



2.3 User Classes and Characteristics

2.3.1 User Classes

User Class	Frequency of Use	Characteristic(s)
Ophthalmologists	High	- Medical professionals specializing in eye care - Familiar with diagnostic tools and medical software
Technicians	Moderate/Medium	- Support staff assisting ophthalmologists - Skilled in operating imaging devices and basic software tools
Patients	Low	- Non-medical users - Interested in understanding their diagnoses and treatment plans
Administrators	Moderate/Medium	- IT professionals responsible for system upkeep

		- Data integrity, and user support

The most important user classes for this product will be Ophthalmologists and Technicians, as they are directly involved in the diagnostic process. Patients and administrators, while essential, are secondary users in terms of system interaction.

2.4 Operating Environment

The system will run on high-performance Linux servers (Ubuntu 20.04+) and support client devices like desktops, laptops, and tablets using Windows 10+, macOS 10.15+, and iOS 13+. It includes Apache for web serving, PostgreSQL for data storage, and AI analysis via PyTorch. The application will use Django on the backend and React on the frontend, with integration options for Patient record systems and compatibility with standard ophthalmic imaging software.

2.5 Design and Implementation Constraints

- **Regulatory Compliance**: Legal constraints (e.g., HIPAA) require strict data privacy and security, limiting options for storage and data sharing.
- **Hardware Limitations**: Limited access to high-performance hardware will restrict model training and deployment, affecting real-time analysis capabilities.
- **Integration with Existing Systems**: Compatibility with EHRs and other healthcare systems will require adherence to specific protocols, reducing flexibility in design.
- **Standardization and Maintenance**: Coding standards and documentation requirements will slow development but ensure maintainability.
- **Language Constraints**: Development is limited to specific languages (e.g., Python), restricting options for performance optimization with other languages.
- **Communication Protocols**: Standard protocols (e.g., REST APIs) will introduce latency, impacting real-time diagnostic performance.

2.6 User Documentation

2.6.1 Documentation Components

User Manuals: Comprehensive guides detailing the system's features, usage instructions, and troubleshooting steps.

Quick Start Guide: A concise guide to help users get started with the system quickly.

2.6.2 Delivery Formats

PDF: User manuals and quick start guides will be available in PDF format for easy download and printing.

2.7 Assumptions and Dependencies

- **Third-Party Components:** It is assumed that reliable third-party libraries and models (pytorch and yolo) for machine learning and image processing will be available and compatible with the project requirements.
- **Operating Environment Stability:** The software is intended to operate in a stable cloud environment with consistent access to necessary computational resources.
- **Data Availability and Quality:** The project assumes access to sufficient, high-quality labeled data for model training.
- **User Hardware and Internet Requirements:** It is assumed that end-users have access to compatible hardware and reliable internet connections.
- **Compatibility with Existing Systems:** The project will integrate with electronic health record (EHR) systems. It is assumed that these systems follow standard data formats and APIs.

3. External Interface Requirements

3.1 User Interfaces

The **User Interface (UI)** for the Pathological Myopia Disease Diagnosis web application will cater to healthcare professionals by providing an intuitive and efficient platform for uploading retinal images, receiving diagnosis results, and navigating patient data. The UI will follow modern standards in healthcare software interfaces, ensuring accessibility, clarity, and quick response times. Below are the details of each component:

3.1.1 Interface Layout

Logical Structure:

- The interface will consist of four main sections: **Image Upload**, **Analysis Dashboard**, Treatment Recommendation and **Results Display**.
- **Image Upload Section**: Allows users to select and upload a fundus retinal image file. This section will include instructions on supported file formats (e.g., JPEG, PNG) and guidelines for image quality to ensure accurate diagnosis.

Navigation:

- The UI will have a left-side navigation bar for easy access to different sections, such as **Home**, **Patient Data**, **Analysis History**, and **Settings**.
- A consistent header with essential functions (e.g., **Help**, **Notifications**, and **Logout**) will appear on each screen.

3.1.2 GUI Standards and Style Guide

- **Color Scheme**: We will use a professional, accessible color palette (e.g., blue tones) that aligns with medical industry standards to instill trust and ensure readability. High contrast colors will be used for important buttons, such as **Submit** and **Download Results**.

- **Typography**: We will use a legible, sans-serif font (e.g., Arial, Helvetica) with a minimum size of 14px for body text and larger fonts for headers.

- **Icons**: Standard icons (such as upload, analysis, and download icons) will be used throughout to enhance recognizability.

GUI Framework:

This application will follow a **Material Design** framework, which supports an intuitive and familiar experience, ensuring that UI elements are responsive and accessible on both desktop and mobile devices.

3.1.3 SCREEN LAYOUT CONSTRAINTS

Responsiveness: The interface will be responsive, adapting to various screen sizes, from desktops in clinical settings to tablets or mobile devices in the field.

Accessibility: All interactive elements will be accessible via keyboard shortcuts for ease of navigation, especially beneficial in clinical environments where rapid data entry is essential.

- **Example Keyboard Shortcuts**:
  - **Ctrl + U**: Opens the **Upload** window.
  - **Alt + A**: Initiates the **Analyze** function.
  - **Esc**: Returns to the **Main Dashboard**.

3.1.4 Standard Functions and Buttons

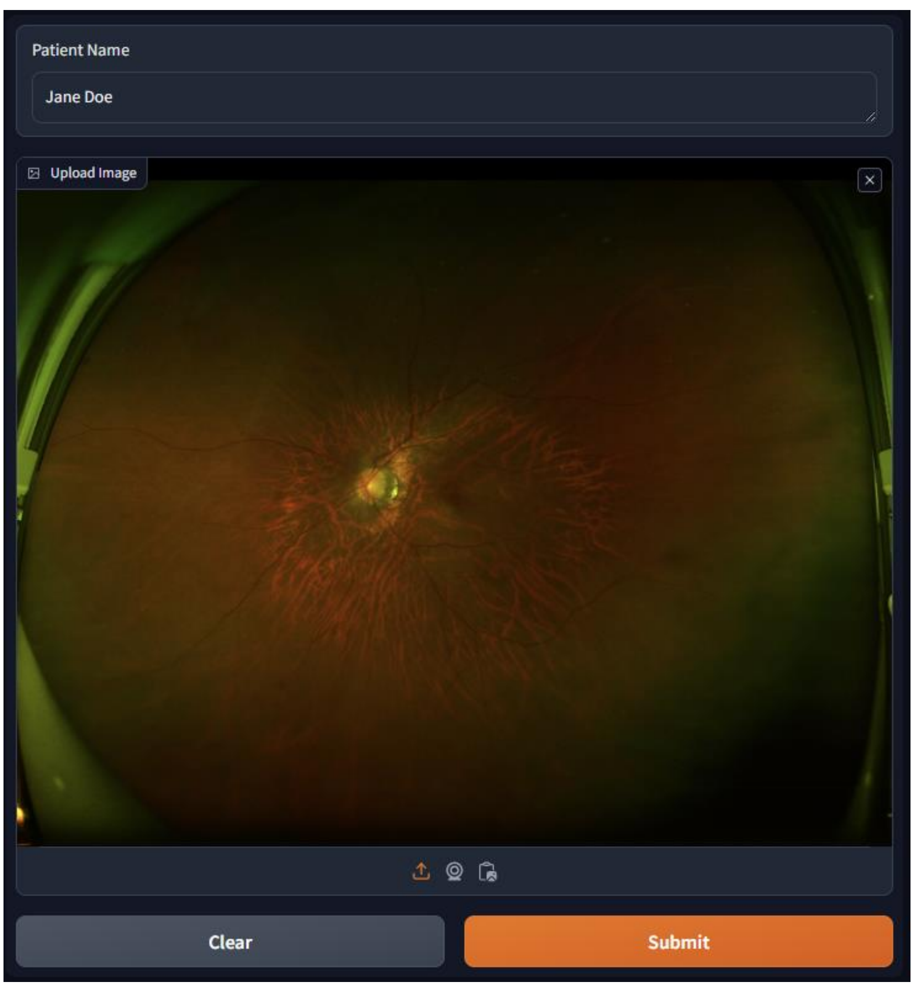
Help: An accessible help button on each screen provides instructions specific to the section the user is viewing.

Error Messages: All error messages will use clear, descriptive language and suggest corrective actions. For example, if the uploaded image is blurry or of incorrect size, the message might read: “Image quality insufficient for analysis. Please upload a clear, high-resolution image.”

Analysis Confirmation: Once the image is uploaded, a confirmation dialog will ask for confirmation before analysis begins, including an estimated analysis time and a reminder to ensure the correct image is selected.

3.1.5 Components Requiring a User Interface

- **Image Upload Component**: Accepts retinal images and provides feedback on the upload status.
- **Real-Time Analysis Component**: Displays a loading indicator while the machine learning model processes the image.
- **Results Display Component**: Shows the diagnostic results, including the probability of pathological myopia, with options to save or print the results.
- **Patient Data Management**: Allows health technicians to review and update patient information related to each diagnostic session.
- **Analysis History and Reports**: Access previous analyses and generate reports for patient history.



WebUI Input

3.2 Hardware Interfaces

The **Pathological Myopia Disease Diagnosis** system relies on specific hardware configurations to support efficient data capture, secure processing, and reliable user interaction. Hardware interfaces for this project will address compatibility, control interactions, and communication protocols to ensure seamless integration of various components.

3.2.1 Supported Device Types

The software will interface with the following hardware devices:

- **Digital Fundus Cameras**:
  - **Purpose**: Used by technicians to capture high-resolution images of the retina, which serve as input for the machine learning model.
  - **Supported Formats**: The system supports JPEG, PNG, and TIFF image formats with resolutions up to 4096 x 3072 pixels for optimal model analysis.
  - **Data Transfer**: Images will be transferred from the camera device to the software interface either through direct upload from local storage or by connecting compatible devices (via USB or wireless protocol).

- **Client Workstations (Desktop Computers or Tablets)**:
  - **Purpose**: Health technicians will use client workstations to access the web application, upload images, view diagnostic results, and manage patient data.
  - **Specifications**: Minimum 4GB RAM, dual-core processor, and a browser that supports HTML5, JavaScript, and CSS3. For faster processing, a GPU-enabled workstation (e.g., NVIDIA GPU support) is recommended to leverage local computation power where possible.
  - **Operating Systems**: Compatible with Windows, macOS, and Linux-based systems to ensure flexibility in different healthcare environments.

- **Servers (Cloud or On-Premises)**:
  - **Purpose**: Hosts the machine learning model and provides backend support for processing and storing patient data and analysis results.
  - **Specifications**: Minimum 8GB RAM, quad-core CPU, and an optional GPU for model inference acceleration. Supports cloud services (AWS, Azure, Google Cloud) with scalable storage and computation.
  - **Communication**: Interfaces with client workstations over HTTPS using RESTful APIs to ensure secure and efficient communication.

- **Mobile Devices**:
  - **Purpose**: For remote access to the analysis system, allowing healthcare providers to view diagnostic results and patient data from mobile devices.
  - **Specifications**: Compatible with iOS and Android operating systems, supporting browsers with HTML5 and JavaScript capabilities.

3.2.2 Data and Control Interactions

- **Data Interactions**:
    - **Image Data**: Digital fundus cameras capture high-resolution retinal images that are uploaded to the application either by connecting the device to a workstation or uploading directly from local storage. Image data will be compressed as needed for efficient transfer without compromising diagnostic quality.
    - **Diagnostic Results**: The server returns diagnosis results to the client device in real-time. The response includes JSON data with key information such as diagnostic classification, probability scores, and recommended actions.

- **Control Interactions**:
    - **Image Capture and Submission**: Health technicians capture images and manually upload them through the web application. For cameras that support direct interfacing, the application may prompt to retrieve images directly from the connected device.
    - **Processing Trigger**: Once an image is uploaded, the backend system initiates model inference automatically. Users can manually trigger re-analysis if additional verification is required.
    - **Error and Status Notifications**: Diagnostic status updates (e.g., "Image processing," "Analysis complete") and error messages are displayed on the client interface. Error conditions (e.g., "Image quality insufficient") are communicated to the user, allowing them to take corrective actions.

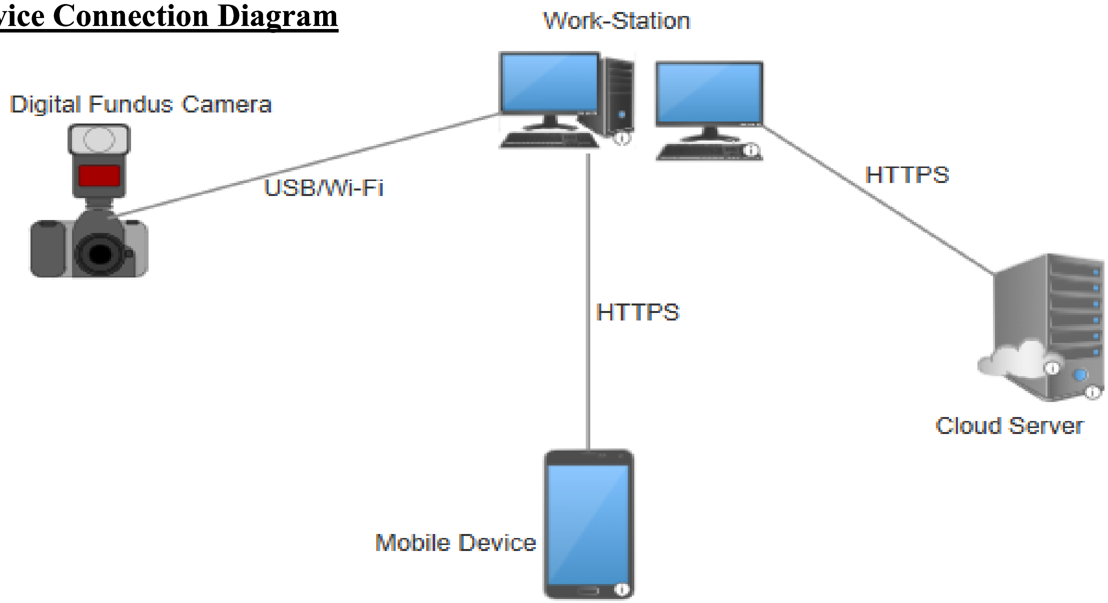
3.2.3 Communication Protocols

- **HTTPS (Hypertext Transfer Protocol Secure)**:
  - **Purpose**: Ensures secure communication between client workstations and the server, protecting patient data and diagnostic information.
  - **Usage**: All data transmissions, including image uploads and result retrieval, will be conducted over HTTPS to comply with data privacy standards, such as HIPAA and GDPR.
  - **USB or Wireless Protocols** (for Image Transfer from Digital Cameras):
  - **USB Protocol**: For devices that connect directly to workstations, USB 3.0 or higher is recommended to allow faster data transfer.
  - **Wireless (e.g., Wi-Fi Direct)**: If supported, wireless protocols can enable direct image upload from the camera to the application, reducing the need for physical connections.

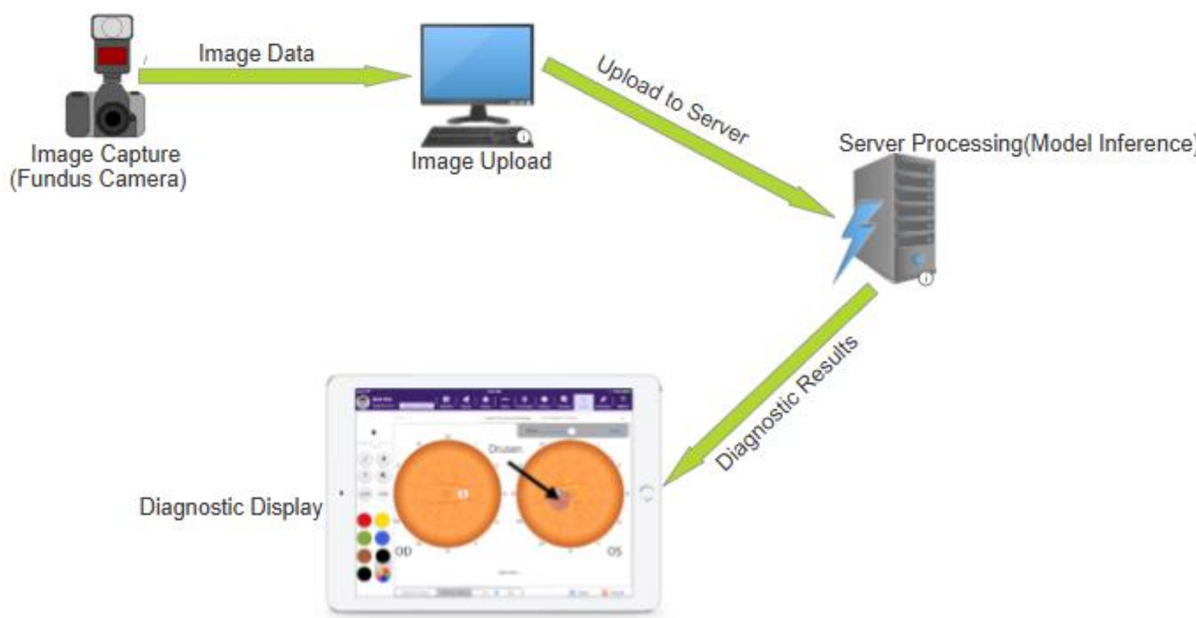
- **RESTful API (Representational State Transfer)**:
  - **Purpose**: Defines interactions between the frontend (web application) and backend (server) for seamless data exchange.
  - **Data Format**: JSON is the primary format for data exchange, supporting requests for image analysis and returning results.

- **WebSockets**:
  - **Purpose**: Enables real-time communication, providing instantaneous feedback to the user for statuses like image processing or analysis completion.
  - **Usage**: WebSockets can be integrated if real-time updates are essential, especially useful for busy clinical environments where prompt notifications are beneficial.

Device Connection Diagram



Data Flow Diagram



3.3 Software Interfaces

The **Pathological Myopia Disease Diagnosis** application will interact with various software components, each playing a critical role in the functionality of the system. This section outlines dependencies on operating systems, databases, external tools, and libraries, as well as how data will flow between these components.

3.3.1 Operating System

The application will be designed to be compatible with multiple operating systems to provide flexibility in deployment:

- **Server Environment**:
  - **Supported OS**: Ubuntu 20.04 LTS (Linux) or Windows Server 2019 for cloud and on-premises hosting.
  - **Role**: Provides the backend infrastructure for model inference, data storage, and secure data exchange.
  - **Communication**: The backend server operates as a RESTful API, handling requests from client applications and returning diagnostic results.

- **Client Environment**:
  - **Supported OS**: Windows 10, macOS (latest versions), and major Linux distributions for desktop clients. Mobile devices running iOS 14 or later and Android 10 or later are also supported.
  - **Role**: Enables health technicians to access the application, upload images, and receive diagnosis results on compatible devices.

3.3.2 Database

- **Database Management System**:
  - **Database**: MongoDB (version 4.4 or later).
  - **Role**: Stores patient data, diagnostic records, and image metadata. Provides a centralized, scalable database for managing both historical and current patient data securely.
  - **Communication**: The application backend uses MongoDB drivers to interact with the database, allowing Create, Read, Update, and Delete (CRUD) operations.
  - **Data Flow**:
    - **Incoming Data**: Patient identifiers, retinal image metadata (e.g., resolution, date captured), and diagnostic results.

        - **Outgoing Data**: Retrieved diagnostic history and patient data for display on the client interface.

    - **Implementation Constraint**: All data access must follow healthcare data protection standards, such as encryption and access control policies.

- **Data Items Shared Across Components**:
    - **Patient Data**: Shared across the backend and front end, including unique patient IDs, demographics, and past diagnoses.
    - **Image Metadata**: Includes image resolution, format, and capture date. This metadata is used by the backend for diagnostic processing and record-keeping.

3.3.3 Machine Learning Model Integration

- **ML Framework**:
    - **Library**: PyTorch (version 1.10 or later).
    - **Purpose**: Hosts the trained machine learning model for pathological myopia detection.
    - **Communication**: The backend server calls PyTorch functions to load the model and process retinal images upon upload.

    - **Data Flow :**
        - **Incoming Data**: Preprocessed image data in tensor format.
        - **Outgoing Data**: Diagnosis results, including probability scores and predicted classifications (e.g., "Pathological Myopia Detected" or "No Pathological Myopia").

    - **Implementation Constraint**: Ensure GPU support for rapid inference if deployed on compatible hardware, enhancing real-time analysis capability.

- **Image Processing Libraries**:
    - **Library**: OpenCV (version 4.5 or later).

- **Purpose**: Processes uploaded retinal images, including resizing, normalization, and format conversion.
- **Communication**: OpenCV functions are called by the backend during preprocessing steps before model inference.
- **Data Flow:**
  - **Incoming Data**: Uploaded image files in various formats (JPEG, PNG).
  - **Outgoing Data**: Preprocessed images converted into a format compatible with the ML model (e.g., normalized tensors).

3.3.4 API Services and Data Communication

- **RESTful API**:
  - **Protocol**: HTTP/HTTPS using JSON format for data exchange.
  - **Purpose**: Manages communication between client applications and the backend server.
  - **Data Flow :**
    - **Incoming Requests**: API calls from the client to the server for image upload, diagnosis, and retrieval of patient data.
    - **Outgoing Responses**: Diagnostic results and status updates sent to the client interface.
  - **Implementation Constraint**: All API endpoints must enforce authentication and authorization to ensure secure access to sensitive data.
- **External APIs**
  - **Cloud Storage API**: Integration with cloud storage services like AWS S3 or Azure Blob Storage, allowing secure storage of large datasets.
  - **Purpose**: Archives retinal images and diagnostic results for long-term storage and retrieval.

- **Communication**: Uses REST API calls over HTTPS to store and retrieve image data as needed.
- **Data Flow :**
  - **Incoming Data**: Processed images and diagnostic results sent for storage.
  - **Outgoing Data**: Retrieved images and records for historical analysis.

3.3.5 User Authentication Service

- **Library**: JWT (JSON Web Tokens) for authentication and role-based access control.
  - **Purpose**: Ensures only authorized users (e.g., health technicians) can access patient data and diagnostic functions.
  - **Communication**: The client application sends a login request, and the server generates a JWT token upon successful authentication.
  - **Data Flow :**
    - **Incoming Data**: Username and password.
    - **Outgoing Data**: Encrypted JWT token for access authorization.
  - **Implementation Constraint**: Ensure tokens are securely stored on the client side and transmitted only over HTTPS.

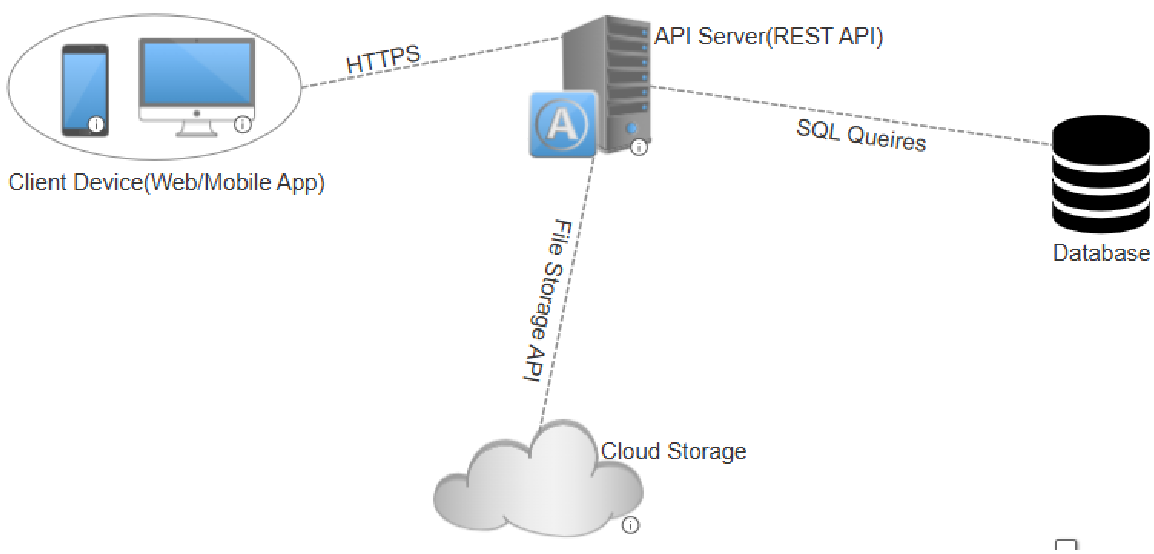
3.3.6 Frontend Libraries and Frameworks

- **React.js** (version 18.0 or later):
  - **Purpose**: Builds the interactive, single-page user interface that allows users to upload images, view diagnostic results, and manage patient records.
  - **Data Flow :**
    - **Incoming Data**: Diagnostic results and status updates from the backend.
    - **Outgoing Data**: User inputs, such as image uploads and search queries.
  - **Implementation Constraint**: Ensure cross-browser compatibility and responsive design to accommodate various devices.

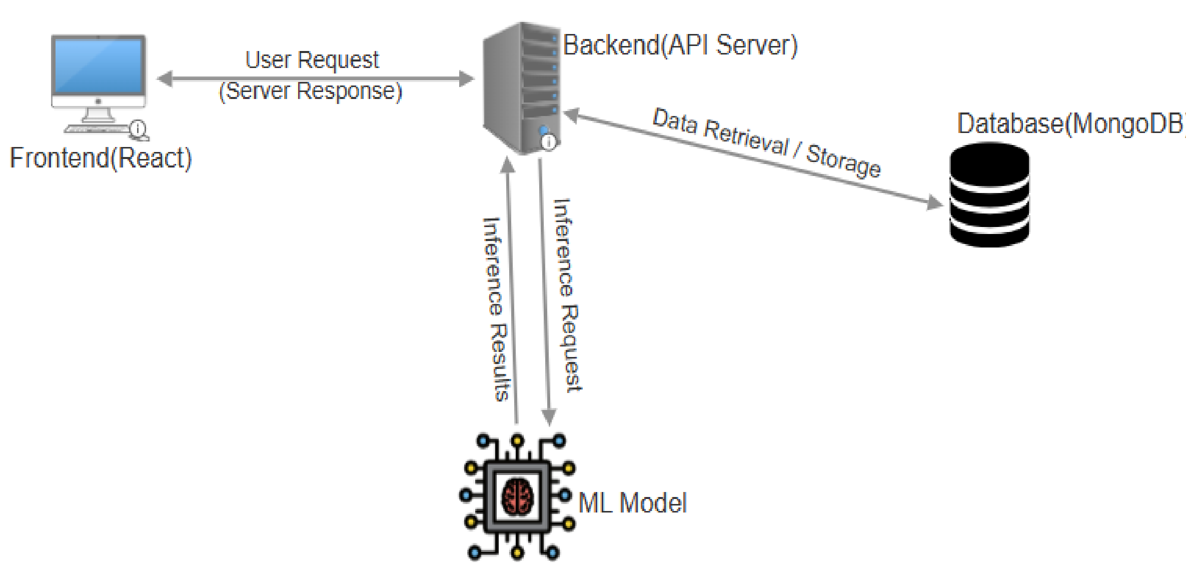
- **Axios**:
  - **Purpose**: Manages HTTP requests from the frontend to the backend API.
  - **Data Flow :**
    - **Incoming Data**: JSON responses from the backend, including diagnostic results and notifications.
    - **Outgoing Data**: JSON-formatted requests to upload images and retrieve data.

  - **Implementation Constraint**: Set request timeouts and error handling to handle network issues efficiently.

System Architecture Diagram



Data Flow Diagram



3.4 Communications Interfaces

The **Pathological Myopia Disease Diagnosis** application requires robust communication functions to enable image uploads, data retrieval, and secure real-time analysis feedback. This section details the protocols, message formats, standards, and security measures that govern these communications.

3.4.1 Web Browser Communication

- **Purpose**: Allows health technicians to access the application interface through any modern web browser.
- **Supported Browsers**: Google Chrome, Mozilla Firefox, Microsoft Edge, and Safari (latest versions) with HTML5 and JavaScript support.
- **Protocol**: **HTTPS** (Hypertext Transfer Protocol Secure) will be used to ensure encrypted data transfer between the client and server.
- **Message Formatting**: All data will be formatted as JSON (JavaScript Object Notation) for API requests and responses, facilitating easy data exchange and parsing by the frontend.

3.4.2 Network Server Communication Protocols

- **HTTPS (HTTP Secure)**:
    - **Purpose**: Ensures encrypted data exchange between client browsers and the backend server, protecting sensitive patient data and diagnostic results.
    - **Implementation**: HTTPS will be enforced across all endpoints, requiring a valid SSL/TLS certificate to establish secure communication.
    - **Security Standards**: Complies with TLS 1.2 or later to meet encryption standards for data protection, ensuring that patient data remains confidential and secure from interception.
    - **Data Transfer Rate**: Standard broadband transfer rates will support the upload and download of images and diagnostic data, but for real-time diagnosis, a minimum upload speed of 5 Mbps is recommended.

- **WebSocket**
    - **Purpose**: Enables real-time, bidirectional communication for immediate feedback on image processing status, diagnostic completion, or error messages.
    - **Protocol**: WebSocket over TLS (WSS), which provides a secure, low-latency channel between the server and client.
    - **Message Formatting**: JSON-formatted messages will be used to transmit processing status updates, allowing technicians to receive notifications without constantly refreshing the interface.
    - **Synchronization Mechanism**: Automatic reconnection will be configured to handle network interruptions and resume communication seamlessly.

3.4.3 Data Transfer Protocols

- **RESTful API**:
  - **Purpose**: Facilitates secure and reliable data exchange between the client-side application and backend services.
  - **Protocol**: HTTP/HTTPS with JSON message format for request and response payloads.
  - **Supported Actions:**
  - **POST**: Used for image upload requests and data submissions.
  - **GET**: Retrieves diagnostic results, patient history, and application status updates.
  - **Security**: Requires token-based authentication (JWT) in HTTP headers for all requests, ensuring that only authenticated users can access sensitive endpoints.

- **FTP/SFTP**
  - **Purpose**: Supports bulk image uploads or batch data synchronization if the application requires integration with external systems or large dataset processing.
  - **Protocol**: **SFTP** (Secure File Transfer Protocol) over SSH is preferred for secure bulk data transfer.
  - **Message Formatting**: Files will be organized with metadata headers that contain patient ID, image capture date, and resolution information for quick identification and processing.

3.4.4 Communication Security and Encryption

- **Token-Based Authentication (JWT)**:
  - **Purpose**: Ensures that only authorized users can access patient data and diagnostic services.
  - **Implementation**: JSON Web Tokens (JWT) will be used for authentication and included in the headers of HTTPS requests, verifying user access without transmitting sensitive data repeatedly.
  - **Encryption Standard**: JWTs will be encrypted using the HMAC SHA-256 algorithm to ensure secure and tamper-proof tokens.

- **Data Encryption**:
  - **In-Transit**: All data exchanged between clients and servers will be encrypted using HTTPS/TLS 1.2 or higher.

  - **At Rest**: Sensitive data, including diagnostic results and patient information, will be encrypted within the database using AES-256 encryption to protect against unauthorized access.

- **Role-Based Access Control (RBAC)**:

  - **Purpose**: Limits access to specific functions and data based on the user's role, ensuring that only authorized personnel (e.g., technicians, administrators) can perform certain actions.
  - **Implementation**: Role-based permissions will be enforced within the API, restricting access to sensitive operations such as data export or administrative functions.

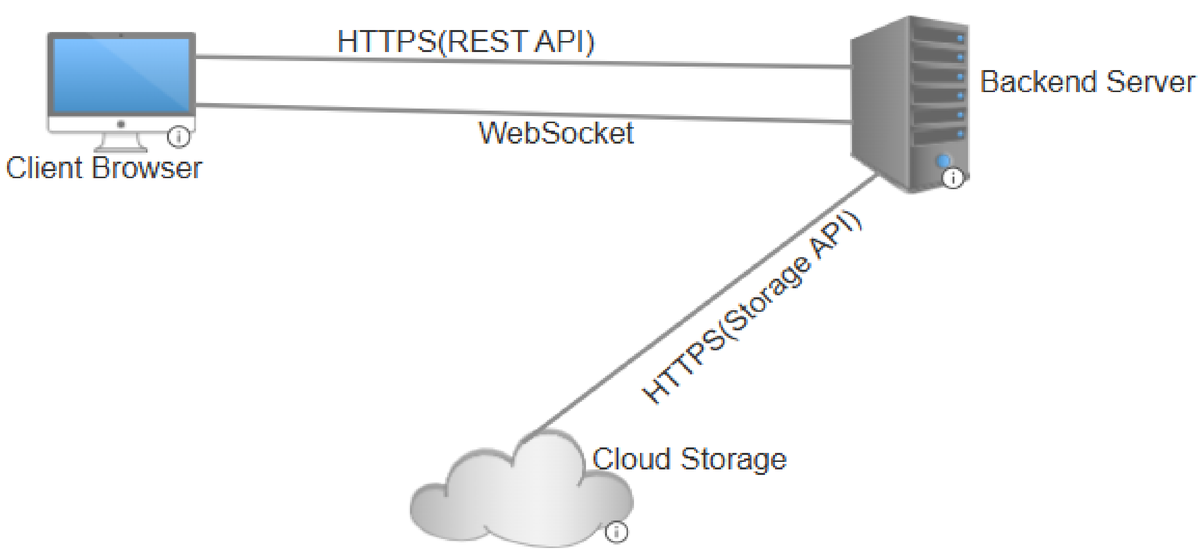
3.4.5 Synchronization Mechanisms

- **Purpose**: Ensures consistency between different system components, such as image uploads and model inference results, by coordinating data transfers and handling reconnections.

- **Implementation:**

  - **Automatic Retries**: Implement retry mechanisms for any failed data transmissions, especially for image uploads, to ensure reliability even with intermittent network issues.
  - **Timestamps**: All diagnostic results and patient records will include timestamps to support data consistency and prevent conflicts, especially for historical record retrieval.
  - **Database Locks**: For operations requiring data consistency (e.g., concurrent access to the same patient record), database locking mechanisms will prevent overwriting data during simultaneous access.

<u>**Network Communication Flow Diagram**</u>



4 System Features

4.1 Image Uploading interface

4.1.1 Description and Priority

Description: Allows specialists to upload fundus images to the system via the web interface

Priority: High

- **Benefit**: 8 - Essential for image acquisition, which initiates the diagnosis workflow.
- **Penalty**: 7 - Without this feature, users cannot submit images for diagnosis.
- **Cost**: 4 - Moderate, depending on compatibility requirements with various devices.
- **Risk**: 5 - Risk is moderate due to varying device formats.

4.1.2 Stimulus/Response Sequences

- **User Action 1**: Specialist connects the imaging device and selects an image to upload.

  System Response: System validates and accepts the image, preparing it for analysis.

- **User Action 2**: Specialist confirms image selection.

  System Response: System confirms successful upload and readiness for analysis.

4.1.3 4.1.3 Functional Requirements

- **REQ-1**: The system shall accept image uploads in standard formats (e.g., JPEG, PNG).
- **REQ-2**: The system shall verify image quality and format compatibility before analysis.
- **REQ-3**: The system shall allow direct upload from connected imaging devices, provided compatibility is established.

4.2 Analysis of Ophthalmic Images

4.2.1 Description and Priority

Description: Analyzes uploaded fundus images using our model to detect signs of pathological myopia.

Priority: High

- **Benefit**: 9 - Core functionality of the system.
- **Penalty**: 8 - Critical as it directly impacts diagnostic capabilities.
- **Cost**: 6 - High due to model development and maintenance.
- **Risk**: 7 - Model accuracy directly affects reliability.

4.2.2 Stimulus/Response Sequences

- **User Action 1**: Specialist will initiate the analysis on the uploaded image.
  System Response: System will process the image, using our machine learning model to analyze for pathological signs and provide observation.

- **User Action 2**: Specialist views the analysis results.
  System Response: System will display diagnostic information, including areas of concern and confidence levels.

4.2.3 Functional Requirements

- **REQ-1**: The system shall process the image using a machine learning model to detect pathological myopia.
- **REQ-2**: The system shall provide a confidence score along with diagnostic results.
- **REQ-3**: The system shall allow re-analysis in case of any manual adjustments to the image.

4.3 Diagnostic Reporting

4.3.1 Description and Priority

Description: Generates a detailed diagnostic report that can be exported as a PDF for the patient and specialist.

Priority: High

- **Benefit**: 8 - Critical for record-keeping and communication with patients.
- **Penalty**: 7 - Necessary for effective information dissemination.
- **Cost**: 4 - Moderate, mainly for formatting and template design.
- **Risk**: 5 - Limited risk; dependent on accurate reporting.

4.3.2 Stimulus/Response Sequences

- **User Action 1**: Specialist requests a report generation.

  System Response: System compiles diagnostic information into a PDF report.

- **User Action 2**: Specialist views and downloads the report.

  System Response: System provides options to download or print the report.

4.3.3 Functional Requirements

- **REQ-1**: The system shall generate a PDF report containing diagnostic details and confidence scores.
- **REQ-2**: The system shall allow specialists to download or print the report.

4.4 Treatment Recommendation

4.4.1 Description and Priority

Description: Suggests potential treatment options based on the diagnostic findings.

Priority: Medium

- **Benefit**: 7 - Enhances diagnostic process by assisting with treatment planning.
- **Penalty**: 6 - Useful but not critical to initial diagnosis.
- **Cost**: 5 - Moderate, depending on recommendation algorithms.
- **Risk**: 6 - Recommendations should be carefully vetted to avoid errors.

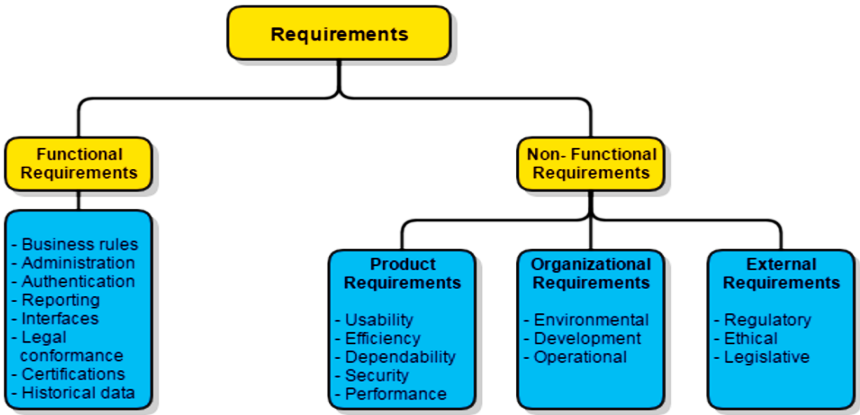
4.4.2 Stimulus/Response Sequences

- **User Action**: Specialist requests treatment recommendations.

  System Response: System displays potential treatments based on the diagnostic outcome.

4.4.3 Functional Requirements

- **REQ-1**: The system shall provide treatment recommendations based on the diagnostic result.
- **REQ-2**: The system shall allow specialists to adjust or ignore recommendations.



5 Other Nonfunctional Requirements

5.1 Performance Requirements

Performance considerations to be incorporated in the System are as Follows:

- **Image Processing Time:** The system will process and analyze each retinal image within 2 seconds to ensure timely diagnosis and feedback for clinicians.
- **Detection Accuracy:** The AI model will be able to achieve a minimum accuracy of 95% in detecting pathological myopia to ensure reliable and trustworthy results.
- **System Uptime:** The system will maintain an uptime of 99.9% to ensure continuous availability for clinicians, minimizing downtime and disruptions.
- **Scalability:** The system will be able to handle up to 10,000 image uploads per day without performance degradation, ensuring it can support high-volume usage in large medical facilities.
- **Latency:** The end-to-end latency from image upload to result display should not exceed 3 seconds to provide a seamless user experience.
- **Data Throughput:** The system will be able to support a data throughput of at least 100 MB/s to efficiently handle large image files and ensure quick processing.
- **Concurrent Users:** The system will support at least 500 concurrent users without performance issues, ensuring multiple clinicians can use the system simultaneously.

- **Error Rate:** The system will have an error rate of less than 0.1% in image processing and detection to maintain high reliability and accuracy.
- **Response Time for API Calls:** The system's API will respond to requests within 200 milliseconds to ensure quick integration and data exchange with other medical systems.
- **Resource Utilization:** The system will utilize not more than 70% of CPU and memory resources during peak usage to ensure there is sufficient capacity for additional load and to prevent system crashes.

5.2 Safety Requirements

The following safety Requirements will be expected of this system:

- **Data Privacy:** Ensure all patient data, including retinal images and diagnostic results, are encrypted both in transit and at rest to prevent unauthorized access and data breaches.
- **User Authentication:** Implement multi-factor authentication (MFA) for all users accessing the system to enhance security and prevent unauthorized use.
- **Data Anonymization:** Anonymize patient data before processing to protect patient identities and comply with privacy regulations such as GDPR and HIPAA.
- **Error Handling:** Implement robust error handling mechanisms to ensure the system can gracefully recover from unexpected failures without compromising data integrity or patient safety.
- **Regular Audits:** Conduct regular security audits and vulnerability assessments to identify and mitigate potential security risks and ensure compliance with industry standards.
- **Access Control:** Implement role-based access control (RBAC) to ensure that only authorized personnel can access sensitive data and system functionalities.
- **Compliance with Medical Standards:** Ensure the system complies with relevant medical standards and regulations, such as ISO 13485 for medical devices and IEC 62304 for medical software.
- **Backup and Recovery:** Implement regular data backup and disaster recovery plans to ensure data can be restored in case of system failure or data loss.
- **User Training:** Provide comprehensive training for clinicians and other users on the safe and effective use of the system, including recognizing and responding to potential safety issues.

- **Incident Reporting:** Establish a clear protocol for reporting and addressing safety incidents, including a system for tracking and resolving reported issues to prevent recurrence.

5.3 Security Requirements

Security requirements to be enforced in the System are as follows:

- **Data Encryption:** All patient data, including retinal images and diagnostic results, will be encrypted using AES-256 encryption both in transit and at rest to ensure data confidentiality and integrity.
- **User Authentication:** Implement multi-factor authentication (MFA) for all users accessing the system to enhance security and prevent unauthorized access.
- **Access Control:** Utilize role-based access control (RBAC) to ensure that only authorized personnel can access specific data and functionalities, minimizing the risk of data breaches.
- **Audit Logging:** Maintain detailed audit logs of all user activities, including data access and modifications, to monitor and trace any suspicious activities or security incidents.
- **Data Anonymization:** Anonymize patient data before processing to protect patient identities and comply with privacy regulations such as GDPR and HIPAA.
- **Regular Security Audits:** Conduct regular security audits and vulnerability assessments to identify and mitigate potential security risks, ensuring compliance with industry standards.
- **Secure API:** Ensure that all API endpoints are secured using HTTPS and require authentication tokens to prevent unauthorized access and data leaks.
- **Incident Response Plan:** Develop and implement a comprehensive incident response plan to quickly address and mitigate the impact of any security breaches or data leaks.
- **Compliance with Regulations:** Ensure the system complies with relevant security and privacy regulations, such as GDPR, HIPAA, and ISO/IEC 27001, to protect patient data and maintain trust.
- **Data Backup and Recovery:** Implement regular data backup and disaster recovery plans to ensure data can be restored in case of system failure or data loss, minimizing downtime and data loss.

5.4 Software Quality Attributes

Below are the Quality Attributes to be respected in this System:

- **Availability:** The system shall have an availability of 99.9%, ensuring it is accessible to clinicians almost all the time, with minimal downtime.
- **Correctness:** The system must correctly identify pathological myopia with a precision of at least 95%, verified through extensive testing with labeled datasets.
- **Flexibility:** The system should be designed to easily incorporate new machine learning models or update existing ones without significant changes to the overall architecture.
- **Interoperability:** The system shall seamlessly integrate with at least 90% of existing electronic health record (EHR) systems used in the target medical facilities, verified through compatibility testing.
- **Maintainability:** The system shall be modular, with a codebase that allows for updates and bug fixes to be implemented within 48 hours, ensuring quick maintenance and improvements.
- **Portability:** The system shall be deployable on various platforms, including cloud-based environments and on-premises servers, with less than 5% performance degradation across different environments.
- **Reliability:** The system shall have a mean time between failures (MTBF) of at least 1,000 hours, ensuring consistent performance and reliability.
- **Reusability:** Components of the system, such as the image preprocessing module and the AI model, shall be reusable in other medical imaging applications, reducing development time for future projects.
- **Robustness:** The system shall handle unexpected inputs, such as corrupted or low-quality images, by providing meaningful error messages and maintaining overall system stability.
- **Usability:** The system shall have a user satisfaction score of at least 85% in usability tests, ensuring that clinicians find the interface intuitive and easy to use.

5.5 Business Rules

Below are the Business rules to be followed along the Software Development Life cycle(SDLC) of this System:

- **Role-Based Access:** Only authorized medical professionals (e.g., ophthalmologists, optometrists) can upload and analyze retinal images. Administrative staff can view but not modify patient data.
- **Data Access:** Clinicians can only access patient data relevant to their cases. Access to other patients' data is restricted to ensure privacy and compliance with regulations.
- **Audit Trails:** All actions performed within the system must be logged, including image uploads, analyses, and data access, to maintain accountability and traceability.
- **Data Retention:** Retinal images and diagnostic results must be retained for a minimum of 5 years, in compliance with medical data retention policies.
- **Patient Consent:** Before uploading retinal images, clinicians must obtain explicit consent from patients, ensuring they are informed about how their data will be used.
- **System Updates:** Only system administrators can perform software updates and maintenance tasks to ensure system integrity and security.
- **Emergency Access:** In case of an emergency, designated senior clinicians can override access restrictions to view all patient data, ensuring timely and effective medical intervention.
- **Data Export:** Only authorized personnel can export patient data for research or reporting purposes, and all exports must be logged and encrypted to ensure data security.
- **Training Requirements:** All users must complete mandatory training on system usage and data privacy policies before being granted access to the system.
- **Incident Reporting:** Any security breaches or system malfunctions must be reported immediately to the IT department, and an incident report must be filed within 24 hours.

6 Other Requirements

- **Database:** We shall use MongoDB for its flexibility and scalability in handling large volumes of unstructured data.
- **Language Support:** The system will support multiple languages, including English, Luganda, and Kiswahili, to cater to a diverse user base.
- **Date and Time Formats**: The system will be able to adapt date and time formats based on the user's locale settings to ensure clarity and usability.
- **Alerts and Notifications:** Alerts and notifications will be set up for critical performance issues to ensure timely intervention and resolution.

6.1 Appendix A: Glossary

AI: Artificial Intelligence

ML: Machine Learning

REQ: Requirement

EHR: Electronic Health Record

MFA: Multi-factor Authentication

PM: Pathologic Myopia

6.2 Appendix B: To Be Determined List

TBD 1: Finalize the performance metrics for the AI model.

TBD 2: Determine the specific legal requirements for data handling.

TBD 3: Confirm the internationalization requirements based on target markets.

TBD 4: References or sources for treatment recommendations given to patients.

TBD 5: Patient identifiers and diagnostic history in the report.